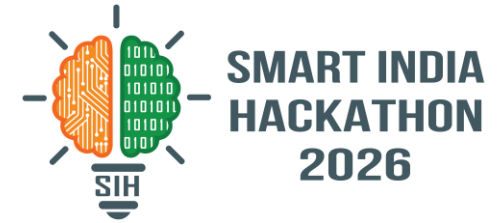


SMART INDIA HACKATHON 2026



TITLE PAGE

ONEPASS

Georeferenced, metrically accurate 3D from a single drone pass

Problem Statement ID

SIH26158

Problem Statement Title

AI-enabled single-pass drone video to a georeferenced, metrically accurate 3D model

Theme

PS Category

Software

Team ID

Team Name (Registered on portal)

fill in

fill in

fill in



IDEA TITLE

Proposed Solution (Describe your Idea/Solution/Prototype)

ONEPASS converts one UAV video plus its GPS log into Earth-referenced, measurable 3D. Telemetry drives the solve instead of being stamped on afterwards.

68	1.57M	287/287	5 of 6
frames kept of 288	fused points, one real run	cameras matched	official formats delivered

Detailed explanation of the proposed solution

1. CAPTURE INTELLIGENCE
- Reads video and GPX / DJI .srt telemetry. Baseline-aware keyframe planning spends the frame budget where parallax is won.
2. METRIC RECONSTRUCTION
- GPS enters bundle adjustment as per-camera priors with per-axis uncertainty; robust Sim(3) into local ENU, then dense stereo.
3. EVIDENCE GATING
- Blur, compression and motion measured as separate gates before matching. 4.85% of pixels vetoed; reprojection error better by 7.8%.
4. GEOREFERENCED DELIVERY
- CRS derived from scene origin; mesh, cloud, DSM, ortho and a walkable browser scene all from the same pass.

How it addresses the problem

Terrain and structures	Poisson mesh + fused dense cloud	Delivered
Facades and rooftops	Oblique/nadir plan, baseline-ranked frames	Mechanism
Roads and infrastructure	Metric DSM, ortho, position table	Delivered
Vegetation and obstacles	Static kept, motion masked	Measured
Textured mesh / point cloud	PLY + LAS now, UV atlas open	Partial

Innovation and uniqueness of the solution

1. GPS AS PRIORS
- Enters bundle adjustment with per-axis uncertainty — never an EPSG stamp pasted onto the output afterwards.
2. BASELINE-AWARE SELECTION
- The one lever that improves geometry and speed at the same time.
3. BLUR AND COMPRESSION SPLIT
- Two failure modes measured and gated separately, not merged into a single score.
4. GEOMETRY THAT CANNOT LIE
- Measured, weak and unobserved surfaces stay separate. No inpainting entry point exists in the code.

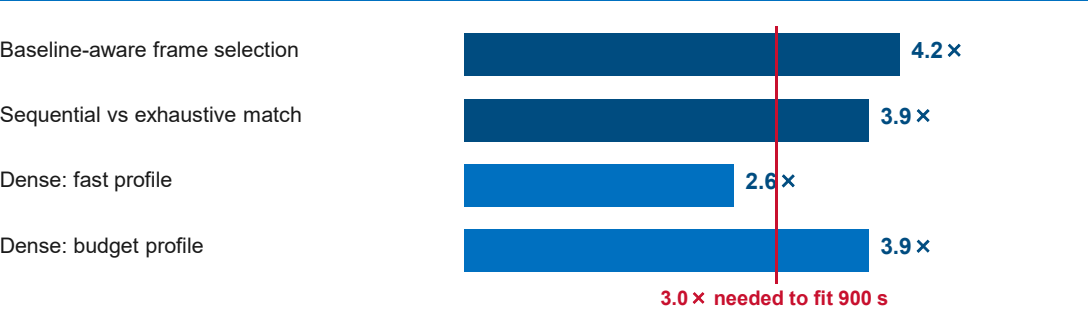
Methodology and process for implementation (Flow Charts/Images/ working prototype)



Technologies to be used (e.g. programming languages, frameworks, hardware)

Language / runtime	Python, CUDA, PyTorch, Docker + conda
Reconstruction core	COLMAP 4.1.1: pose-prior mapper, PatchMatch, Poisson
Geometry / alignment	Robust Sim(3) + RANSAC, WGS84 to local ENU
Frame intelligence	ffmpeg decode; Laplacian, Tenengrad, blocking scores
Telemetry	GPX, DJI .srt, Pilot CSV, contract CSV readers
Output products	PLY, LAS, GeoTIFF, glTF/GLB, FBX, WGS84 table
Interface / QA	PlayCanvas viewer, Next.js console, headless-Chrome walk

Measured speed-up on one RTX 3050 6 GB laptop (×)



First bar is estimated from measured per-image rates; the other three are measured stage times.

4:30

demo video

11/11

CPU suites green

151

survey tests

1,405 MiB

peak VRAM

65.3 m

walked, 0 falls

8/8

cells at 2 m

Working prototype: repository with continuous integration and a 4 minute 30 second recorded demo.

Analysis of the feasibility of the idea

Parameter	Official target	Position today
Reconstruction	3D mesh / point cloud	Delivered
Processing time	< 15 min per 10-min video	Not met
Spatial accuracy	≤ 1 m	Unproven
Coverage	Entire visible scene	Partial
Output formats	6 formats specified	5 of 6
Visualisation	Web or desktop viewer	Delivered

Against the brief's eight key challenges: 1 fulfilled with measurement, 5 mechanism-complete awaiting data, 1 partial by design, 1 not met on this hardware.

Potential challenges and risks

Dense stereo needs ~3 × more speed at a 270-frame aerial budget

Consumer GNSS is 1–3 m and worse vertically

Occluded faces cannot be imaged from a single path

Exports never read by a third-party reader

Official evaluation criteria — weight and evidence

Accuracy		30
Mechanism complete, unproven on qualifying data		
Completeness		20
Observability measured, 8 of 8 cells at 2 m		
Speed		20
2.6–4.2 × levers, time gate still unmet		
Innovation		15
Delivered and distinct from survey-first pipelines		
Scalability		10
Runs on one 6 GB laptop, peak 1,405 MiB		
Usability		5
Walkable viewer, 65.3 m, 16 of 16 waypoints		

Accuracy, completeness and speed carry 70 of 100 points — and all three sit inside the reconstruction core. Usability is worth 5. That weighting is why the build order went to geometry before the viewer.

Strategies for overcoming these challenges

- Three measured levers behind one profile flag plus windowed submaps; one 10-minute flight run twice proves it
- GPS enters as uncertainty-weighted priors with takeoff-anchored heights; RTK/PPK kept as a separate declared track
- The hole is measured and published with its excluded area; generative inpainting deliberately absent
- Add GDAL, PDAL and laspy validation gates to CI before the qualifying flight

Potential impact on the target audience

One engine turns the footage a mission already captures into a measurable world — offline, on hardware a team already carries.

BORDER AND PERIMETER

Hostile terrain mapped without a survey crew entering it

DISASTER FIRST RESPONSE

Damage model before the first team lands

FACILITIES AND TELEMETRY

Existing sites digitised from footage already flown

MISSION REHEARSAL

Walkable scene of the objective area before approach

Military reconnaissance

AVAILABLE TODAY — a 65.3 m walkable terrain scene, 16 of 16 waypoints, 0 falls, with position, ground, obstacle and flyable data.

ONE BUILD AWAY — enemy indicators (helmet, vehicle, tent, gun) and full mission-rehearsal data that flying alone cannot supply.

Urban planning

AVAILABLE TODAY — georeferenced buildings, roads, vegetation and utilities from real scenes; 5 of 6 official formats in one run.

ONE BUILD AWAY — semantic layers and a text legend.

Supporting: disaster response, AI-in-the-loop flight optimisation, digital twins, historical preservation, forestry and agriculture, inspection.

Benefits of the solution (social, economic, environmental, etc.)

SOCIAL

First damage model in minutes for search and rescue, with no cell tower and no cloud. The terrain of a landslide or a flood is walked remotely before anyone walks it for real.

ECONOMIC

One flight instead of a survey sortie and a ground-control crew. Developed on a 6 GB laptop peaking at 1,405 MiB — no workstation to buy, no ground station to establish.

STRATEGIC AND ENVIRONMENTAL

Runs fully offline for contested or air-gapped sites. Licence-clean and reproducible from one command. Fewer repeat sorties over the same ground.

Target audience: field engineers, disaster-response and border-mapping agencies, survey and construction firms, facility and autonomous-operations teams.

Details / Links of the reference and research work

RECONSTRUCTION AND MAPPING	FEED-FORWARD GEOMETRY AND TRACKING	RENDERING, PHYSICS AND EVALUATION	THIS SUBMISSION'S EVIDENCE TRAIL
COLMAP SfM and MVS colmap.github.io	DPVO deep patch visual odometry github.com/princeton-vl/DPVO	gsplat Gaussian splatting stack docs.gsplat.studio	Problem statement and desired output SIH26158 PDF, pp. 38–40
OpenDroneMap photogrammetry github.com/OpenDroneMap/ODM	Pi3 equivariant reconstruction github.com/yyfz/Pi3	MuJoCo physics engine mujoco.org	Evaluation and improvement report 09_SIH26158_Evaluation...2026-09-22
Single-pass UAV trajectory planning github.com/ch1bo/drone-reconstruction	Depth-Anything-3 github.com/ByteDance-Seed/Depth-Anything-3	evo trajectory evaluation github.com/MichaelGrupp/evo	Evidence map, wiring ledger, findings docs/readiness/04, 02, 01
	MapAnything metric 3D mapping github.com/facebookresearch/map-anything	RGB-D and completion datasets dornhelge.github.io	Prototype and demo github.com/krisgarg25/Drone_Phone_video...
	VGGT-Long online reconstruction github.com/DengKaiCQ/VGGT-Long		

EVIDENCE ACCOUNTABILITY

Measured figures are stage times, frame counts, point counts and errors read from machine-generated reports. Anything not yet proven is labelled a target and stated as unproven. The 1.14 m hold-out figure is reported against a drifting phone inertial reference and is presented as reference-dependent rather than as a pass. No total evaluation score is shown, because only the accuracy and completeness terms could be computed.